

Three simultaneous polynomial-size dice in fully fair families

Complete proof; three independent reviews passed

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Theorem 1. *There are absolute constants $C, c > 0$ such that for every sufficiently large n there is a fully permutation-fair family of n finite dice in which three specified dice have at most Cn^c faces each. All faces on every die are equiprobable and all labels are globally distinct. The other $n - 3$ dice may have one common finite face count; no polynomial bound on that common count is asserted.*

The exponent is deliberately not optimized. The proof controls one common denominator for all initial cell means, then performs one final local insertion. It does not by itself establish a statement for four or for an arbitrary fixed number of small dice.

Lemma 1 (One further variable from controlled cell means). *Suppose rational densities f_1, f_2 are supported on complementary pure cells forming a rational partition of $[0, 1]$. Assume:*

1. *the number of cells, inverse cell lengths, inverse positive cell masses, and polynomial piece degrees are bounded by a polynomial in m ;*
2. *the conditional density on each cell, rescaled to $[0, 1]$, is bounded above and below by absolute positive constants;*
3. *there is one polynomial integer D such that every endpoint, every positive cell mass, and every unnormalized first cell moment belongs to $D^{-1}\mathbb{Z}$.*

Then there are three rational piecewise polynomial densities $\tilde{f}_1, \tilde{f}_2, \tilde{f}_3$ on pairwise disjoint pure cells whose ordered polynomial moments through total degree m coincide with those of independent variables having densities $f_1, f_2, 1$. Their cell masses belong respectively to $N_1^{-1}\mathbb{Z}, N_2^{-1}\mathbb{Z}, N_3^{-1}\mathbb{Z}$ for polynomial integers N_i . Their rescaled conditional density bounds are absolute, and their polynomial piece degrees are polynomial in m . There is no required bound on the new cell endpoint or higher-moment denominators, nor on the number of new cells beyond finiteness.

Proof. Consider an original cell $I = [a, b]$, length ℓ , occupied by old label $i \in \{1, 2\}$. Put

$$\gamma = \int_I f_i(t) dt, \quad z = \int_I t f_i(t) dt.$$

After rescaling I to $[0, 1]$, its old conditional density is a rational polynomial w with mean

$$\mu = \frac{z - a\gamma}{\gamma\ell}.$$

Choose polynomial integers $N_1 = N_2 = N$ and $N_3 = T$, both multiples of D , sufficiently large, and prescribe the local counts

$$K_I = N\gamma, \quad L_I = T\ell.$$

These are positive integers. Moreover

$$K_I L_I \mu = NT(z - a\gamma) \in \mathbb{Z}, \tag{1}$$

since $D^2(z - a\gamma)$ is integral.

Apply the theorem in `../size_bounds/generalized_local_two_variable_completion.tex` on every cell, with its old density w as the theorem's B variable and its uniform variable as the theorem's A variable. It is applicable for all integers

$$K_I \geq C(m + \deg w + 2)^4, \quad L_I \geq C[K_I(m + \deg w + 2)]^{160}$$

satisfying (1). First take N , a multiple of D , as a sufficiently large fixed polynomial in m to satisfy every first inequality. Then take T , another multiple of D , as a sufficiently large fixed polynomial to satisfy every second inequality. The uniform bounds in the hypotheses, the polynomial lower bounds on γ, ℓ , and the common D make these simultaneous choices possible. No least common multiple of independently produced cell denominators is taken.

Rescale each local output back to I . Multiply the old density by its original cell mass γ , and the new density by the original uniform mass ℓ . Each local old box has global mass $\gamma/K_I = 1/N$. Each local new gap has mass an integer times $\ell/L_I = 1/T$. Assemble the old pieces over their respective original cells to obtain $\tilde{f}_1, \tilde{f}_2$, and assemble the new pieces over all original cells to obtain $\tilde{f}_3$. Each integrates to one. Their supports are pairwise pure, every coefficient and endpoint is rational, and their claimed common mass denominators are N, N, T . The local theorem gives the uniform conditional density bounds and polynomial piece degrees.

It remains to check the global moments. Partition an ordered integral according to the three original cells occupied by its variables. The first two variables never occupy the same original cell. If all three cells are distinct, their relative order is fixed and the integrand is an ordinary polynomial; the local marginal moment identities preserve it. If the third variable shares one of the first two cells, the ordering constraint on that pair is the only nontrivial local chamber condition. The local two-variable theorem preserves all its polynomial moments through degree m , and the other old variable is in a different cell with fixed relative order. Expand the polynomial in the other variable and use its preserved marginal moments. These cases exhaust the partition. The original cell masses are unchanged, so summing proves every asserted ordered moment identity. $\square$

Proof of Theorem 1. Write $m = n - 3$. Use `two_pure_densities_with_rational_means.tex` with parameter $m + 1$ to obtain f_1, f_2 such that these variables and $m + 1$ independent uniforms are fully permutation-fair. Its cell bounds and single polynomial denominator verify every hypothesis of Lemma 1. Separate one of those uniforms as the third specified variable and apply that lemma through degree m . Conditional on the values of the three specified variables, a fixed full order of the remaining m uniforms has probability, on its specified chamber,

$$\prod_{j=0}^3 \frac{(t_{j+1} - t_j)^{r_j}}{r_j!}, \quad t_0 = 0, \quad t_4 = 1, \quad \sum_j r_j = m.$$

This is a polynomial of total degree m . The lemma preserves its integral in each chamber. Thus the three new rational pure densities and m uniforms are still fully permutation-fair.

Let their polynomial mass denominators be N_1, N_2, N_3 . Choose a common polynomial integer H sufficiently large. On each pure cell belonging to label i , its mass is an integer a/N_i . Replace its conditional density by an equal-weight real quadrature using exactly Ha nodes, exact through degree at least $2m + 2$. Every sufficiently large number of nodes is permitted, uniformly in the interval and cell: the separated quadrature lemma in `../size_bounds/generalized_local_two_variable_completion.tex` applies after rescaling. Its hypotheses are exactly the absolute upper and lower density bounds and the polynomial degree bound established above. Taking its exactness degree to be the larger of $2m + 2$ and one more than that piece degree, one polynomial choice of H works on all cells. It gives interior, pairwise distinct nodes directly. Increase H also to ensure $H \geq m^2$.

The resulting specified dice have exactly HN_i equiprobable nodes, polynomial in m . On each product of pure cells their relative order is fixed, so replacing each density by its conditional quadrature preserves every degree- m order kernel. They remain fully fair with the m uniforms. Each pure cell supplies a cluster containing at least m^2 distinct nodes. Shrink each cell to a closed subinterval containing its finitely many nodes in its interior, giving disjoint cluster closures. The cluster moments through degree m are rational, because they equal moments of a rational polynomial density on an interval with rational endpoints, multiplied by its rational cell mass.

All hypotheses of the reviewed finiteization theorem `../size_bounds/clustered_multi_exceptional_lifting.tex` now hold for $k = 3$. It gives a completely finite fair family without changing the three specified counts HN_i . The remaining m dice have a common finite count. Replacing the global face ordering by successive integer labels makes all labels distinct. Finally, the three counts HN, HN, HT can be equalized to $H \operatorname{lcm}(N, T) \leq HNT$ by uniform face replication within each specified die. Place each replica in a sufficiently small disjoint neighborhood of its old label; every relative order is unchanged. The common specified count is still polynomial. This proves the theorem. $\square$